

GAS & OIL FIELD SERVICES COMPANY

Stores Operations, Stock Control and ERP Workflow Report

Integrated Operating Model for Ahmadi, Coiled Tubing, Drilling and PCP Stores

Prepared for	GOFSCO Management
Document purpose	Stores governance, ERP design and implementation
Locations	Ahmadi Coiled Tubing Drilling PCP
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Core principle: One item master, one controlled transaction process and one source of truth across all four stores. No physical movement is considered complete until it is posted in the ERP.

Document Control

Field	Requirement
Document owner	Stores Supervisor / Company Assets Supervisor
Process owners	Stores, Supply Chain, Finance, Operations and HSE
Approver	Executive Management / authorized delegated authority
Review cycle	At least annually and whenever the ERP, organization, facility or legal requirements change
Controlled copies	ERP document repository; obsolete versions must be archived and marked superseded
Related records	PO, delivery note, GRN, MIV, MRV, transfer order, count sheet, adjustment approval, SDS, credit note and disposal approval

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1. Executive Summary

GOFSCO should operate its four stores as one integrated inventory network rather than four independent warehouses. Ahmadi, Coiled Tubing, Drilling and PCP will retain their physical responsibility for custody and service to operations, while item definitions, transaction rules, approval limits, valuation, document control and performance reporting remain standardized.

The recommended ERP model uses a single item master and separates stock by site, warehouse, zone, rack, shelf and bin. Each quantity also carries a stock status such as available, reserved, inspection, quarantine, damaged, expired, customer-owned, vendor-owned or scrap. This structure allows management to see exactly what GOFSCO owns, where it is, whether it can be used, which job or department needs it and what action is pending.

Management outcome: The system should reduce stockouts, emergency purchases, duplicate buying, undocumented issues, expired chemicals, unexplained variances and slow-moving inventory while improving job readiness, cost allocation, auditability and cash control.

1.1 Current-state problems requiring correction

The current Sage 300 process depends heavily on users navigating separate screens, manually entering information from hardcopy forms, printing reports and then exporting or reorganizing the results in Excel. The screenshots and process notes supplied for item creation, purchase orders, receiving, internal usage/MIV, physical inventory, MIV reporting, MRV reporting and aged inventory confirm that the present process is functional but fragmented and outdated for a four-store operation.

Observed problem	Operational impact	Required ERP response
Hardcopy-to-soft-copy transcription	Typing handwritten MIV/MRV, delivery and count information creates item-code, quantity, date, location, UOM, cost-center and job-reference errors.	Digital request/approval at source; barcode scanning; mandatory fields; validation and electronic acknowledgement.
Delayed system entry	Physical stock moves before the system is updated, so the balance shown is temporarily or permanently incorrect.	Point-of-activity mobile posting; same-shift control; ageing alerts for unposted or pending transactions.
Repeated manual entry	The same PO, item, quantity and reference information may be re-entered across purchase, receipt, MIV/MRV and reporting steps.	Document flow that carries approved data forward automatically and prevents unnecessary re-keying.
Outdated, multi-screen navigation	Users must know exact Sage menu paths and complete several legacy screens, increasing training time and dependence on individual knowledge.	Role-based modern workspace, guided tasks, search, favorites and mobile workflows.
Excel used to complete reporting	Inventory worksheets, MIV/MRV and aged-inventory reports are exported and manually organized, creating version-control and formula/filter risks.	Live dashboards and scheduled reports using governed ERP data; export remains optional, not a processing requirement.
Limited workflow visibility	The user cannot easily see whether a request is waiting for approval, picking, receipt, inspection, return, credit note or posting.	End-to-end status tracking, owner, timestamp, service deadline and escalation.
Weak bin/status detail	A location code alone may not distinguish rack/bin or whether stock is available, reserved, quarantined, damaged, expired or in transit.	Warehouse-zone-rack-shelf-bin hierarchy plus a separate controlled stock-status dimension.
Inconsistent master-data completion	Item screens allow many fields, but completion and naming depend on the user; duplicate or incomplete codes can be created.	Central item request, duplicate search, mandatory attributes, naming convention and maker-checker approval.

Observed problem	Operational impact	Required ERP response
Limited lot/expiry use	Chemical batch, expiry and SDS data may not be systematically enforced at receipt and issue.	Mandatory lot/expiry/SDS fields by category, FEFO allocation, warnings and hard issue blocks.
Physical inventory is paper/Excel-oriented	Worksheets may show default on-hand quantities and are printed/exported for modification, weakening blind-count discipline and audit control.	Mobile blind counts, controlled freeze/cut-off, automated recount and electronic variance approval.
Reports are retrospective	Management learns about shortages, aged stock and errors after reports are generated and manually reviewed.	Real-time exception alerts for low stock, excess, ageing, expiry, pending inspection, in-transit and variance.
Knowledge concentrated in individuals	Step-by-step knowledge is stored in screenshots or personal experience instead of embedded controls.	Standard workflow, contextual guidance, training records, SOP links and role-based permissions.

Primary control weakness: The present process separates the physical event, paper evidence, Sage posting and Excel reporting into different activities. Every handoff creates a chance for delay, omission, duplicate entry or mismatch. The future workflow must capture the transaction once at source and reuse the same approved data throughout.

1.2 Current Sage 300 process assessment

Current activity shown	Current handling	Future-state requirement
Create item code	Navigate to I/C Items and Price Lists, open Items, manually complete fields and save.	Guided item request with duplicate detection, category-specific mandatory fields and approval before activation.
Create purchase order	Open P/O Transactions and Purchase Order Entry, then fill order, vendor, locations and line information.	Approved requisition automatically converts to PO with inherited item, quantity, cost center, job and delivery data.
Receive material	Open Receipt Entry, enter PO and quantities/costs, match against invoice, post and print MRV/receipt form.	Scan PO/delivery, record accepted quantity and condition, attach evidence, route inspection, post GRN and trigger three-way match.
Issue material	Open I/C Transactions and Internal Usage, manually enter description/reference/user, line item, location, quantity and UOM, post and print MIV.	Approved electronic request creates pick task; scan item/bin; system posts issue and captures receiver acknowledgement.
Inventory worksheet	Generate worksheet, select ranges/options, print or export to Excel and modify/organize.	System-generated blind mobile count by zone with controlled recount, investigation and adjustment approval.
MIV/MRV reports	Run specific stock-control reports, export to Excel and organize as needed.	Searchable transaction dashboard with drill-down to digital MIV/MRV, attachments, department, job and approval history.
Aged inventory	Generate department-wise output, export to Excel and organize.	Automated ageing/FSN dashboard with last movement, months of cover, value, owner and disposition workflow.

1.3 Business consequences of retaining the current method

- Physical and system quantities can differ because paperwork and posting do not occur at the same moment.

- Errors can pass from hardcopy into Sage and then into Excel without automated validation.
- Stores spends time preparing and cleaning reports instead of controlling stock and supporting operations.
- Management cannot depend on a single live view of available, reserved, quarantined, expired and in-transit stock across all four stores.
- Urgent requirements may cause verbal issues or late paperwork, increasing the backlog and weakening job-cost accuracy.
- The company may repurchase material already held at another store because network availability is difficult to see.
- Chemical expiry, returns, vendor credit notes and obsolete stock can remain open without clear ownership or escalation.
- Audit evidence is spread across paper, Sage and separate Excel files, making transaction reconstruction slower and less reliable.

1.4 Target operating principles

- ERP first: every approved stock movement is recorded at the point of activity using barcode or QR scanning wherever practical.
- No document, no movement: goods may not be received, issued, returned, transferred, adjusted or disposed without the required reference and approval.
- One item, one code: duplicate item records are prevented through centralized item-master governance.
- Location accuracy: every item is assigned to a controlled location down to bin level, with clear labels and no informal storage areas.
- Stock status is separate from physical location: quarantined or damaged material remains visible but cannot be issued.
- Safety overrides convenience: chemical compatibility, SDS requirements, ventilation, containment and emergency arrangements govern storage.
- Segregation of duties: requesting, approving, receiving, recording, counting and adjusting are separated to the extent practical.
- Exception-driven management: dashboards highlight shortages, overdue approvals, expiry, excess stock, negative stock attempts and count variances.

2. Scope, Objectives and Boundaries

2.1 Scope

This report covers all inventory and store-related activities at the following GOFSCO locations:

Store	Primary role	Typical inventory
Ahmadi Store	Central/common store and network control point	Common consumables, PPE, tools, filters, oils, spares, general materials, selected chemicals and shared stock
Coiled Tubing Store	Dedicated operational support for CT activities	CT consumables, pressure-control and equipment spares, hoses, seals, lubricants, chemicals and job-specific material
Drilling Store	Dedicated operational support for drilling/workover activities	Rig and drilling spares, mechanical/electrical consumables, PPE, filters, oils, tools and slow-moving strategic spares
PCP Store	Dedicated support for artificial-lift/PCP operations	PCP equipment and spares, elastomers, drive-related items, accessories, tools, consumables and job-specific stock

2.2 Objectives

- Provide reliable stock availability for operations without carrying uncontrolled excess.
- Create seamless, auditable workflows from demand planning through purchase, receipt, storage, issue, return and disposal.
- Establish clear ownership and accountability at each store.
- Provide management with real-time visibility of stock quantity, value, condition, expiry, demand and movement.
- Integrate Stores with Operations, Procurement, Finance, HSE, Maintenance and Asset Management.
- Create a scalable foundation for ERP implementation, barcode scanning and future automation.

2.3 Boundaries

Inventory consists of consumables, spare parts, chemicals and controlled operational material intended for issue or use. Capital assets and serialized equipment remain linked to the Asset Register, although their receipt, custody transfer and storage location may pass through the Stores process. Customer-owned, rental, consignment and vendor-owned material must be clearly identified and must never be valued as GOFSCO-owned inventory.

3. Governance and Organization

3.1 Governance structure

Role	Core accountability
Executive Management	Approve policy, disposal, material write-offs, major stock provisions and authority limits.
VP Operations / Operations Management	Set operational priorities, approve critical demand and resolve cross-department stock conflicts.
Stores Supervisor	Own policy, network stock control, master data coordination, reporting, cycle counts and continuous improvement across all four stores.
Storekeeper / Location Custodian	Control access, receive, inspect, label, put away, issue, return, count and maintain housekeeping at the assigned store.
Department / Operations Manager	Own demand forecast, technical specification, min/max recommendation, reservation priority and confirmation of obsolete stock.
Supply Chain / Procurement	Source approved requirements, maintain PO accuracy, coordinate delivery and vendor performance.
Finance	Control valuation, three-way matching, credit notes, provisions, reconciliation and approved write-offs.
HSE	Approve chemical storage controls, emergency arrangements, inspections, training and incident response.
Quality / Technical Inspector	Perform or coordinate technical inspection where required and release or reject material.
ERP Administrator / IT	Maintain roles, workflows, interfaces, backups, audit logs, reports, devices and system availability.

3.2 Segregation of duties

Activity	Initiator	Approver / verifier	System posting
Purchase request	Authorized requester	Department manager / budget authority	Supply Chain / ERP workflow
Receipt	Storekeeper	PO and delivery verification; technical inspector when applicable	Authorized receiving role
Issue	Requester	Department/job approver	Storekeeper
Return	Returning employee/department	Storekeeper condition verification	Storekeeper
Transfer	Requesting or sending store	Stores Supervisor / authorized manager	Sending and receiving custodians
Adjustment	Storekeeper / count team	Independent reviewer and authority based on value	Restricted inventory-control role
Disposal	Stores + department owner	HSE/Finance/Management as applicable	Restricted role after physical disposal evidence

Access rule: Store access shall be limited to authorized personnel. Visitors, technicians and drivers must be escorted. Keys and system access must not be shared.

4. Four-Store Network and Location Design

4.1 Network operating model

The four stores should be configured as separate ERP warehouses under the same company and inventory organization. Stock is visible across the network, but physical custody and transactions remain location-specific. Ahmadi should act as the central control and common-material hub; operational stores should hold activity-specific and job-critical stock.

ERP code	Warehouse	Recommended responsibility	Replenishment model
AHM	Ahmadi	Common stock, central oversight, shared buffer and selected reserve stock	Forecast + min/max + network consolidation
CTU	Coiled Tubing	CT-dedicated consumables, spares and chemicals	Job plan + equipment PM + critical min/max
DRL	Drilling	Drilling/workover consumables and strategic rig spares	Rig plan + usage history + criticality
PCP	PCP	PCP operational material and job-specific equipment/spares	Job schedule + BOM/kit + min/max

4.2 Location hierarchy and coding

Every storage position must have a unique hierarchical code:

Location hierarchy

Site	Warehouse	Zone	Rack	Shelf	Bin
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Example: AHM-GEN-R03-S02-B04. Chemical areas use a dedicated prefix, such as CTU-CHM-COR-R01-S01-B02. Outdoor laydown areas, cages, containers, cold storage, quarantine and scrap zones are coded in the same manner.

Zone code	Purpose	Issue status
REC	Receiving and document verification	Blocked
INS	Technical/quality inspection	Blocked pending release
AVL	Approved available stock	Issuable
RES	Reserved or job-kitted stock	Restricted to reservation
RET	Returned material awaiting assessment	Blocked
QUA	Quarantine/nonconforming/unknown material	Blocked
DMG	Damaged material	Blocked
EXP	Expired chemical/material	Blocked
SCR	Approved scrap awaiting disposal	Blocked
CON	Consignment/vendor-owned stock	Controlled by ownership terms

4.3 Physical layout standards

- Clearly marked receiving, inspection, quarantine, returns, approved stock, job staging, chemicals, cylinders, hazardous waste and scrap zones.
- Racks and shelves rated for the intended load; heavy items stored at low level; aisles and emergency routes kept clear.
- Weather-sensitive materials stored indoors or in suitable sealed/conditioned containers.
- High-value, attractive or controlled items stored in locked cages with restricted access and transaction review.
- Floor locations marked for large items with maximum stacking height and load limits.
- Each bin displays location code, item code, description, unit of measure and barcode/QR code; quantity is displayed only if operationally useful.
- No unidentified items. Unknown material is tagged and moved to quarantine until identified and dispositioned.
- A permanent staging zone is not used as overflow storage; aged staging items appear on an exception report.

5. Item Master and Data Governance

5.1 Item creation standard

The item master is the foundation of ERP accuracy. A controlled Item Creation/Change Request must be approved before a new code is created. The master-data team checks for duplicates using description, manufacturer, part number, specification and alternate numbers.

Mandatory field	Requirement
Item code	Unique, non-intelligent sequential code or governed category code; never reused.
Short/long description	Standard noun-modifier format with size, rating, material and key technical attributes.
Category/subcategory	Consistent reporting and approval routing.
Manufacturer / part number	Captured separately; alternate part numbers maintained where technically approved.
Base unit of measure	Single inventory UOM such as EA, PAIR, L, KG, DRUM or M; purchasing conversions defined.
Ownership	GOFSCO, customer, vendor/consignment, rental or temporary custody.
Stock class	Consumable, spare, chemical, tool, repairable, serialized item, capital asset or non-stock service.
Criticality	Critical, essential or routine, with technical justification.
Planning parameters	Lead time, safety stock, reorder point, minimum, maximum, order multiple and preferred warehouse.
Control attributes	Lot/batch, serial, expiry, shelf life, SDS, inspection, calibration or warranty as applicable.
Finance attributes	Inventory account, cost center rules, valuation method and tax/category data.
Storage requirements	Temperature, humidity, indoor/outdoor, compatibility group, hazardous classification and special handling.
Status	Draft, active, blocked, phase-out or obsolete.

5.2 Description convention

Recommended format: NOUN, TYPE, KEY SPECIFICATION, SIZE/CAPACITY, MATERIAL/RATING, MANUFACTURER, PART NUMBER. Example: FILTER, ENGINE OIL, HINO 500, OEM/APPROVED EQUIVALENT, P/N XXXXX.

5.3 Units of measure

Purchasing and stock units must not be confused. If gloves are purchased by carton but issued by pair, the ERP must contain a controlled conversion (for example, 1 carton = 120 pairs). Conversions are approved and locked; free-text UOMs are prohibited.

6. Inventory Classification and Planning

6.1 Classification

Method	Purpose	Recommended use
ABC	Classifies annual consumption value	A: tight control/frequent counts; B: normal control; C: simplified control.
Criticality	Measures operational consequence of unavailability	Critical items may require stock despite low consumption.

Method	Purpose	Recommended use
FSN	Fast, slow and non-moving based on issue frequency	Identifies excess and obsolete stock.
VED	Vital, essential and desirable	Supports operational prioritization.
Shelf-life	Separates expiry-controlled items	FEFO, alerts and restricted receipt rules.
Repairability	Separates repairable/rotatable items	Tracks condition, repair order and custody.

6.2 Minimum, maximum and reorder rules

Planning parameters must be set per item and warehouse. A critical CT seal may need minimum stock at CTU, while common gloves may be centrally held at Ahmadi. Parameters are reviewed quarterly and whenever activity, supplier lead time, consumption or equipment population changes.

Parameter	Recommended calculation / rule
Average demand	Validated average consumption over a representative period, adjusted for confirmed work plans.
Safety stock	Demand variability and supply uncertainty during lead time, increased for critical items.
Reorder point	Expected demand during lead time + safety stock.
Maximum stock	Reorder point + planned order cycle quantity, constrained by shelf life, space and budget.
Order quantity	Maximum minus available projected stock, rounded to supplier pack/order multiple.
Available projected stock	On hand + open PO + approved inbound transfer - reservations - expected demand.
Emergency level	A management alert when available stock falls below the defined operational survival quantity.

6.3 Demand planning

- Operations submits a rolling 12-month demand forecast, with a firm near-term window and indicative later months.
- Job-specific material is linked to job number, customer, well/rig/equipment and planned date.
- Maintenance spares are linked to equipment BOM and preventive-maintenance schedules.
- Stores compares demand against on-hand, reserved, open purchase orders, transfers and historical usage.
- Supply Chain receives a consolidated replenishment proposal to prevent duplicated orders across departments.
- Forecast accuracy and emergency purchase frequency are reviewed monthly with Operations.

6.4 Excess, slow-moving and obsolete stock

The ERP should calculate months since last issue and months of cover. A quarterly review board determines whether material will be retained, transferred, returned to vendor, used as an approved substitute, sold, repaired, donated where permitted or scrapped. No item is written off only because it has not moved; technical criticality and equipment life must be considered.

7. End-to-End ERP Workflows

7.1 Demand to replenishment

Standard replenishment

Demand/Min-Max Alert	Stock Check	Transfer Check	PR Approval	PO	Expected Receipt
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- The requester selects an existing item code, required quantity, need-by date, warehouse, department, cost center and job/equipment reference.
- The ERP displays network availability, reservations, open purchase orders and approved substitutes before a PR is submitted.
- Internal transfer is considered before external purchase where stock is genuinely excess at another store.
- Budget and technical approval routes depend on category, value, urgency and criticality.
- Emergency requests require justification and are reported separately for root-cause review.

7.2 Receiving

Receipt workflow

Arrival Booking	PO/Delivery Check	Unload & Count	Inspection	GRN	Label & Put-Away
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Step	Control
Delivery scheduling	Vendor/Procurement confirms PO, delivery location, date, vehicle and special handling. Unapproved chemical deliveries are not accepted.
Gate/arrival	Delivery note and PO are verified; material is directed to receiving, not directly to shelves or user departments.
Physical check	Count, packaging, visible damage, manufacturer, part number, batch/serial and expiry are checked.
Documentation	PO, delivery note, certificates, SDS, test/calibration documents and warranty are attached to the ERP receipt.
Inspection status	Material requiring technical or quality inspection is received into inspection status and cannot be issued.
GRN	Posted only for the quantity actually accepted. Partial receipts and rejected quantities are recorded accurately.
Put-away	ERP proposes a suitable bin; storekeeper scans item and bin, then confirms quantity.
Discrepancy	Over/short/damaged/wrong material is recorded and Procurement is notified; it remains segregated pending resolution.

Invoice control: Finance processes supplier invoices only against the approved PO, posted GRN and supplier invoice. Exceptions are routed for resolution rather than forced through the system.

7.3 Issue to department, job or employee

Issue workflow

Material Request	Approval	Reservation/Pick List	Pick & Verify	MIV Posting	Handover
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- The request must identify warehouse, item, quantity, cost center, department, job/work order and requested date.
- ERP reserves available stock only after approval. Critical shortages are allocated by Operations Management.
- Storekeeper follows the system pick list using FIFO or FEFO and scans the item, lot/batch/serial and bin.
- The receiver confirms quantity and custody electronically or by signed MIV attached to the ERP.
- Negative stock, backdated issues and verbal undocumented withdrawals are prohibited.
- Unused job kits are returned promptly using the return workflow; they are not kept indefinitely in vehicles or work areas.

7.4 Returns from operations

Internal return

Return Request	Physical Handover	Condition Check	MRV	Disposition	Put-Away/Repair/Scrap
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Condition	ERP disposition
Unused, sealed and serviceable	Return to available stock after verification.
Opened but usable	Return only when identity, condition and shelf life can be assured; otherwise quarantine.
Repairable	Move to repairable status/location and create repair/work order.
Damaged/nonconforming	Move to quarantine or damaged status and raise nonconformance/disposition request.
Expired or contaminated	Move to blocked hazardous-waste/expired zone for HSE-approved disposal.
Unknown identity	Quarantine until positively identified; never merge with known stock.

7.5 Return to vendor and credit note

Vendor return

Return Approval	Vendor RMA/Acceptance	Dispatch Note	ERP Vendor Return	Credit Note	Finance Reconciliation
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No vendor return is closed until Finance confirms the credit note or approved replacement. Returned quantities leave usable stock immediately and remain traceable through the vendor-return reference.

7.6 Inter-store transfer

Inter-store transfer

Transfer Request	Approval	Pick & Dispatch	In-Transit Stock	Receive & Verify	Put-Away
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- A transfer order identifies sending store, receiving store, item, quantity, reason and required date.
- The sending store posts shipment to an in-transit virtual location; stock does not appear as available at both stores.
- A dispatch note and custody acknowledgement travel with the shipment.
- The receiving store independently counts and scans the material. Differences create a transfer discrepancy case.
- Urgent transfers may use an expedited route but still require ERP posting and post-approval within the defined limit.

7.7 Stock adjustment

Adjustments are exception transactions, not normal corrections. The ERP requires item, location, quantity, value, reason code, supporting evidence, count reference and approval. Users who count or hold custody should not approve their own material adjustment.

Reason code	Example evidence
COUNT_GAIN / COUNT_LOSS	Approved count sheet and recount result
UOM_ERROR	Master-data review and corrected conversion
DAMAGE	Photograph, incident/nonconformance record and disposition
EXPIRY	Batch/expiry report and HSE disposition
DATA_MIGRATION	Approved migration reconciliation
SCRAP_WRITE_OFF	CEO/management approval and disposal certificate/evidence

7.8 Scrap and disposal

Disposal workflow

Identify & Segregate	Technical Review	Value/HSE Review	Management Approval	Physical Disposal	ERP Write-Off
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Scrap reports should list item code, description, quantity, condition, location, ownership, book/stock value where applicable, reason, photographs and recommended method. Disposal must preserve environmental, data-security and customer/property obligations. ERP write-off occurs only after approval and evidence of physical disposal.

8. Chemical Storage and Control

Mandatory rule: Chemicals may be delivered only against an approved PO and agreed delivery schedule. The receiving store must verify the current SDS, label, container condition, batch and expiry before acceptance. Vendor delivery directly into storage without Stores approval is prohibited.

8.1 Chemical master data

- Product and chemical name, manufacturer, product code and approved use.

- Hazard classification and pictograms as stated in the current SDS.
- SDS revision/date and attached electronic copy accessible to workers.
- Compatibility/storage group, temperature range, ventilation and light/moisture restrictions.
- Container size, base UOM, batch/lot, manufacture date, expiry/retest date and shelf life.
- Required PPE, spill response, firefighting information and disposal route.
- Ownership, customer/job restriction and regulatory/transport information where applicable.

8.2 Storage design

Control area	Minimum requirement
Compatibility	Segregate strictly by SDS and approved compatibility assessment; do not rely only on product name or color.
Secondary containment	Bunded/contained storage sized for credible spills and suitable for the chemical; keep drains protected.
Ventilation	Provide natural or mechanical ventilation appropriate to the hazard assessment.
Temperature	Monitor where specified; alerts and documented action for excursions.
Ignition control	Control heat, sparks, smoking and electrical suitability for flammable areas.
Shelving	Chemical-resistant, stable and load-rated; liquids below powders where leakage could cause reaction/contamination.
Labels	Original label preserved; decanted containers require compliant identity and hazard labeling.
Emergency equipment	Spill kits selected for the chemical, eyewash/shower where required, fire equipment, emergency contacts and clear access.
Security	Restricted access; high-hazard chemicals locked and issued only to authorized/trained users.
Housekeeping	No damaged containers, residues or mixed waste; routine leak and corrosion inspection.

8.3 Chemical receiving and release

Chemical receipt

Approved PO	SDS/Label Check	Container & Batch Check	Expiry Acceptance	GRN in Inspection	Release & FEFO Put-Away
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- Define minimum remaining shelf life at receipt by product category; short-dated material requires written operational acceptance.
- Damaged, leaking, unlabelled or incompatible deliveries are isolated safely and rejected or quarantined.
- Samples and unknown chemicals are treated as controlled chemicals, not placed with normal stock.
- Batch certificates or certificates of analysis are captured when required by Operations/customer specification.

8.4 Chemical issue, return and expiry

- Use FEFO based on actual expiry/retest date; the ERP blocks expired batches.
- Issue by batch to job/cost center and record the receiving authorized person.
- Part-used returns are accepted only if the original container, seal/closure, label, content integrity and storage history are acceptable.

- The ERP triggers alerts at agreed intervals, for example 180, 90, 60 and 30 days before expiry.
- Expired, contaminated or leaking chemicals are moved to a labelled hazardous-waste/blocked area pending HSE-approved disposal.
- Vendor returns require return authorization and Finance credit-note closure.

8.5 HSE inspections and emergency readiness

HSE and Stores should jointly inspect chemical areas using a standard checklist covering SDS availability, labels, segregation, containment, ventilation, temperature records, spill kits, emergency equipment, housekeeping, container condition, expiry and training. Incidents and near misses are recorded, investigated and linked to corrective actions.

9. Stock Control and Inventory Accuracy

9.1 Transaction discipline

- Transactions are posted in real time at the correct warehouse, bin, status, lot/serial, cost center and job.
- Backdating is restricted to authorized Finance/Inventory Control users and requires a reason.
- ERP must prevent negative stock and issuing blocked, expired or reserved material.
- Manual spreadsheets may support analysis but must not become an alternative stock ledger.
- System downtime transactions use pre-numbered controlled forms and are entered immediately after restoration with supervisor review.

9.2 Cycle counting

Stock class	Minimum frequency	Method
A-value / high-risk / controlled	Monthly or more frequently	Blind count by independent counter; immediate recount of variance.
B-value	Quarterly	Rotating cycle count.
C-value	Semiannual	Rotating or zone count.
Critical operational items	Monthly / before major job	Quantity and condition verification.
Chemicals	Monthly	Quantity, batch, expiry, container condition and SDS check.
Full inventory	Annually	Planned count with movement freeze or controlled cut-off.

9.3 Count workflow

Cycle count

Freeze/Control Movements	Blind Count	Recount Variance	Investigate	Approve Adjustment	Close Corrective Action
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Count sheets must not show expected quantities to the first counter. Variances are investigated for timing, UOM, wrong bin, unposted issue/return, duplicate item, transfer, damage or theft. Repeated causes result in corrective action, not repeated adjustment.

9.4 Annual stocktake

- Approved count plan, teams, areas, cut-off date, tagging rules and system freeze/controlled movement process.
- All receipts, issues, returns and transfers posted or clearly isolated before cut-off.
- Independent count teams and signed count sheets or controlled mobile count tasks.
- Reconciliation by item, warehouse, status, quantity and value with management approval.
- Finance sign-off and audit trail retained according to document-retention rules.

10. Daily, Weekly, Monthly and Annual Tasks

Frequency	Storekeeper / location tasks	Stores Supervisor / network tasks
Every shift / daily	Open/close security check; receive and issue; scan put-away; review staging; housekeeping; chemical visual check; resolve pending transactions; handover log.	Review critical shortages, blocked receipts, overdue requests, negative-stock attempts, transfer discrepancies and urgent operational requirements.
Weekly	Location/bin audit; check unidentified items; review open returns and job kits; inspect racks, spill kits and quarantine; verify documentation completeness.	Review open PO/receipts, slow approvals, stockouts, emergency issues, inter-store opportunities and data-quality exceptions.
Monthly	Cycle counts; chemical expiry/condition check; damaged/obsolete identification; equipment and safety checklist.	Inventory accuracy report; min/max exceptions; ageing/FSN; KPI meeting; Finance reconciliation; HSE chemical review.
Quarterly	Deep housekeeping and zone review; verify labels and location maps.	Reclassify ABC/FSN; review min/max and lead times; obsolete board; access review; vendor/department performance review.
Annually	Full stocktake support; training refresh; emergency drill participation.	Policy review; full stocktake; master-data cleanse; authority/access recertification; insurance/valuation support; ERP continuity test.

11. ERP Functional Requirements

11.1 Core modules

Module	Required capability
Master data	Governed item, vendor, location, UOM, category, lot/serial, chemical and approval data.
Inventory	Multi-warehouse/bin, stock status, lot/serial, reservations, transfers, counts, adjustments and valuation.
Procurement	PR, approval, RFQ, PO, expected delivery, partial receipt, vendor return and performance.
Warehouse mobile	Barcode/QR receiving, put-away, picking, issue, return, transfer and counting.
Maintenance/Assets	Equipment BOM, spare linkage, work-order demand, serialized asset receipt and custody.
Finance	Inventory accounts, cost centers, job costing, GRNI/accrual, three-way match, credit notes and write-offs.

Module	Required capability
HSE/Quality	SDS, inspection status, nonconformance, expiry, certificates, hazardous storage and disposal records.
Document management	Attach approvals, delivery notes, certificates, photos, SDS, signed MIV/MRV and disposal evidence.
Analytics	Dashboards, alerts, scheduled reports, ageing, accuracy, service level, value and exception trends.

11.2 Required system controls

- Role-based access by store, transaction and approval limit; maker-checker controls for sensitive activity.
- Audit log showing who created, approved, changed or posted each record and when.
- No deletion of posted transactions; corrections occur through controlled reversal and reposting.
- Duplicate item detection and mandatory master-data fields.
- Prevention of negative stock, expired issue, blocked-stock issue and invalid UOM conversion.
- Configurable approvals by value, category, urgency, department and exception type.
- Electronic signatures or authenticated approvals with delegated authority and absence cover.
- Automated alerts for low stock, overdue PO, pending inspection, expiry, excessive stock, aged staging and count variance.
- Interface controls and reconciliation between inventory, procurement, finance, maintenance and assets.
- Backup, disaster recovery, cybersecurity, device management and tested offline/downtime procedure.

11.3 ERP status model

Status	Available to promise?	Permitted action
Available	Yes	Reserve, transfer or issue
Reserved	Only for linked demand	Pick/issue to reservation or authorized release
Inspection	No	Inspect and release/reject
Quarantine	No	Investigate and disposition
Damaged	No	Repair, vendor return or scrap
Expired	No	Return/dispose through HSE process
In transit	No	Receive or investigate transfer discrepancy
Consignment	According to contract	Issue and trigger ownership/settlement rules

11.4 Dashboards

- Stock by location, category, ownership, status and value.
- Critical items below minimum and days until projected stockout.
- Open requests, reservations, purchase orders, overdue deliveries and pending receipts.
- Pending inspection, quarantine, returns, vendor credit notes and transfers in transit.
- Chemicals by batch, expiry horizon and storage location.
- Slow/non-moving and excess stock with last issue date and months of cover.
- Inventory accuracy, count completion and adjustment value by store.

- Emergency issues/purchases, service level and department forecast accuracy.

12. Reporting and KPIs

KPI	Definition / target direction
Inventory record accuracy	Correct count lines ÷ total counted lines; increase toward an agreed target, with stricter target for A/critical items.
Stock availability/service level	Approved demand fulfilled in full by required date ÷ total approved demand; increase.
Stockout rate	Demand lines unavailable when required ÷ total demand lines; decrease.
Emergency purchase rate	Emergency PR/PO lines ÷ total PR/PO lines; decrease.
Inventory turnover	Annualized consumption/cost of issues ÷ average inventory value; optimize by category.
Slow/non-moving value	Value exceeding defined no-movement thresholds; decrease after technical review.
Expiry loss	Value of material expired/write-off; decrease.
Receiving turnaround	Time from accepted delivery to available put-away; decrease without bypassing inspection.
Issue turnaround	Time from approved request to handover; decrease.
Cycle-count completion	Completed planned counts ÷ scheduled counts; target 100%.
Adjustment value	Absolute value of adjustments as percentage of inventory value; decrease.
Document completeness	Transactions with all mandatory evidence ÷ sampled transactions; target 100%.
Transfer discrepancy rate	Transfer lines with quantity/condition discrepancy ÷ transfer lines; decrease.
Chemical compliance	Passed chemical control checks ÷ checks performed; target 100% for critical controls.

Targets should be baselined during the first three months of reliable ERP operation. Management should avoid setting artificial targets before transaction and master-data accuracy are stabilized.

13. Internal Controls, Audit and Document Retention

- Sequential ERP transaction numbers and controlled forms for downtime.
- Approval matrix maintained by Finance/Management and reflected in ERP workflow.
- Quarterly access review; immediate removal or change upon transfer/termination.
- Monthly reconciliation of inventory subledger to the general ledger.
- Sample audits of PO-GRN-invoice, MIV, MRV, transfer, adjustment, chemical receipt and disposal transactions.
- Retention periods aligned with company policy, customer contracts, financial requirements and applicable Kuwait requirements.
- Photos and scanned forms must be readable, linked to the transaction and protected from unauthorized alteration.
- Exception logs show overdue approvals, reversed/backdated transactions, manual overrides, repeated variances and blocked-stock activity.

Audit expectation: The ERP should show the complete story of an item: why it was bought, when it arrived, who accepted it, where it was stored, who received it, which job used it, whether it returned and how any remaining quantity was disposed.

14. Roles and RACI Summary

Process	Stores	Operations	Supply Chain	Finance	HSE/Quality	Management
Demand forecast	C	R/A	C	C	I	I
Item master request	R	R/C	C	C	C	A by governance
Min/max review	R	R/C	C	C	C	A for major policy
Purchase	C	R request	R/A	C	C	I/A by limit
Receipt/put-away	R/A custody	C inspect	C	I	C/R inspect	I
Issue/return	R	R request/receive	I	I	C	A by authority
Transfer	R	C	I	I	I	A by authority
Count/adjustment	R	C	I	C/A control	C	A by value
Chemical control	R	C	C	I	A	I
Obsolete/disposal	R	R technical	C	C	C/A HSE	A

R = Responsible, A = Accountable, C = Consulted, I = Informed.

15. Implementation Roadmap

Phase	Indicative duration	Key deliverables
1. Mobilize and govern	2-3 weeks	Sponsor, process owners, project team, scope, authority matrix, issue log and implementation calendar.
2. Physical and data baseline	4-6 weeks	Full count, location mapping, ownership/status segregation, chemical inventory, item duplicate review and opening reconciliation.
3. Design and configure	6-10 weeks	Warehouse/location structure, master data, workflows, roles, approvals, forms, reports, integrations and barcode standards.
4. Cleanse and migrate	4-8 weeks	Approved item master, UOM conversions, lot/serial/expiry, opening balances, open PO and reservation data.
5. Pilot Ahmadi	4-6 weeks	End-to-end pilot, devices, training, transaction testing, count reconciliation and issue resolution.
6. Roll out CTU, Drilling and PCP	6-10 weeks	Staged go-live by store, cutover count, on-site support and daily controls.
7. Stabilize and optimize	12 weeks	KPI baseline, parameter tuning, dashboard adoption, obsolete review and continuous improvement backlog.

15.1 Critical cutover requirements

- No migration of unidentified or duplicate stock into available inventory.
- Opening quantity and value reconciled and approved by Stores and Finance.
- All locations labelled and physically ready before loading stock.
- Users trained and access tested before go-live.
- Open POs, reservations, returns, transfers and inspection items agreed at cutover.
- Daily command-center review during stabilization, with ownership and due date for every issue.
- Legacy spreadsheets changed to read-only archive after acceptance; ERP becomes the official ledger.

15.2 Training

Training must be role-based and practical. Storekeepers complete receiving, put-away, picking, issue, return, transfer, counting and chemical scenarios. Requesters and approvers practice correct item selection, job/cost-center coding and mobile approvals. Finance, HSE and administrators receive dedicated control and reporting training. Competence is confirmed by supervised transactions, not attendance alone.

16. Risk Register and Mitigation

Risk	Likely effect	Primary mitigation
Poor item data / duplicates	False stockouts, duplicate buying and unreliable reports	Governed item creation, cleanse and duplicate check.
Transactions posted late	ERP differs from physical stock	Mobile scanning, same-shift posting and exception dashboard.
Unauthorized access	Loss, safety exposure or manipulation	Physical and role-based access with periodic review.
Incorrect UOM conversion	Large quantity/value errors	Central approval and locked conversions.
Chemical incompatibility	Fire, reaction, exposure or environmental harm	SDS-based segregation and HSE approval/inspection.
Overstocking	Cash tied up, expiry and space shortage	Min/max review, network visibility and excess-stock board.
Understocking	Job delay and emergency buying	Criticality, demand forecast, safety stock and lead-time review.
Unclosed returns/transfers	Stock disappears into staging or transit	Ageing alerts, custody confirmation and escalation.
ERP downtime	Unrecorded movements	Tested continuity forms, recovery entry and reconciliation.
Weak change management	Users maintain parallel spreadsheets and bypass controls	Sponsor enforcement, training, support and KPI ownership.

17. Recommended Management Decisions

1. Approve the four-store integrated operating model and designate the ERP as the sole official inventory ledger.
2. Appoint process owners for Stores, Operations, Supply Chain, Finance, HSE and ERP administration.
3. Approve the warehouse/location coding standard and physically label all areas before migration.

4. Authorize a full physical inventory and master-data cleansing exercise across Ahmadi, CTU, Drilling and PCP.
5. Approve mandatory PO/Stores authorization for all deliveries, particularly chemicals.
6. Approve the transaction and approval matrix, including adjustment, emergency issue, transfer and disposal limits.
7. Approve barcode/QR devices, printers, labels and connectivity required for point-of-activity posting.
8. Require monthly Stores performance review and quarterly excess/obsolete stock review.

18. Conclusion

A seamless GOFSCO stores operation depends less on the ERP brand than on disciplined data, clear ownership and mandatory workflows. The four stores should operate as one network with standardized rules while preserving local custody and operational responsiveness. When every item has one identity, every quantity has one known location and status, and every movement has an approved transaction, the ERP can provide reliable stock availability, cost visibility and audit control.

The recommended approach should be implemented in controlled phases, beginning with physical and data accuracy. This prevents the company from transferring existing inconsistencies into a new system and creates a practical foundation for automation, performance management and future growth.

Appendix A - Required ERP Transaction Types

Transaction	Minimum references / data
Purchase receipt / GRN	PO, delivery note, supplier, item, quantity, UOM, warehouse/bin, lot/serial/expiry, inspection status and attachments
Material issue / MIV	Approved request, receiver, department, cost center, job/work order, item, quantity, lot/serial and handover confirmation
Material return / MRV	Original issue where available, returning party, condition, quantity, disposition and receiving confirmation
Inter-store transfer	Sending/receiving warehouses, dispatch/receipt custodians, in-transit status, item, quantity and discrepancy
Vendor return	Original PO/GRN, vendor authorization, reason, dispatch proof, replacement/credit-note reference
Stock adjustment	Count/reference, reason, quantity/value, investigation, evidence and approvals
Status change	From/to status, reason, inspection/disposition and approver
Scrap/write-off	Approved scrap report, photos, quantity/value, disposal method/date and evidence

Appendix B - Mandatory Forms and Records

- Item Creation/Change Request
- Material Requirement / Purchase Requisition
- Delivery Booking and Receiving Checklist
- Goods Receipt Note (GRN)
- Inspection/Release or Rejection Record
- Material Issue Voucher (MIV)
- Material Return Voucher (MRV)
- Inter-Store Transfer Order and Dispatch/Receipt Note
- Vendor Return and Credit Note Follow-Up
- Cycle Count / Annual Count Sheet and Variance Investigation
- Stock Adjustment Request
- Chemical Receiving and Monthly Inspection Checklist
- Nonconformance / Quarantine Disposition
- Scrap Findings and Disposal Approval Report with photographs
- System Downtime Transaction Form and Recovery Reconciliation

Appendix C - Minimum Location Readiness Checklist

Control	Ahmadi	CTU	Drilling	PCP
Warehouse, zone, rack, shelf and bin labels installed	☐	☐	☐	☐
Receiving and inspection zones marked	☐	☐	☐	☐
Quarantine, damaged, return and scrap zones marked	☐	☐	☐	☐
Chemical areas assessed and compatibility segregation approved	☐	☐	☐	☐

Control	Ahmadi	CTU	Drilling	PCP
Racks/load limits and emergency aisles verified	☐	☐	☐	☐
Barcode/QR devices, printer and connectivity tested	☐	☐	☐	☐
Full count and ownership/status verification completed	☐	☐	☐	☐
Store users trained and access tested	☐	☐	☐	☐
Opening balance reconciled with Finance	☐	☐	☐	☐
Go-live approval signed	☐	☐	☐	☐